Pandas Complete Roadmap (Beginner → Advanced) with Descriptions

---

1️⃣ DataFrame & Series Creation

Method	Description
`pd.Series(data)`	1 D labeled array তৈরি করে
`pd.DataFrame(data)`	2 D table (rows & columns) তৈরি
`pd.read_csv(filename)`	CSV file থেকে DataFrame load
`pd.read_excel(filename)`	Excel file থেকে DataFrame load
`pd.read_json(filename)`	JSON file থেকে DataFrame load
`pd.read_sql(query, conn)`	SQL থেকে DataFrame load

2️⃣ Data Inspection

Method	Description
`df.head(n)`	প্রথম n row দেখায়
`df.tail(n)`	শেষ n row দেখায়
`df.info()`	column & dtype summary
`df.describe()`	numerical summary (mean, std, min, max)
`df.shape`	DataFrame dimensions
`df.columns`	column names
`df.index`	row index

3️⃣ Selection & Indexing

Method	Description
`df['col']`	column selection
`df[['col1','col2']]`	multiple columns
`df.loc[row_index, col_name]`	label-based selection
`df.iloc[row_index, col_index]`	position-based selection
`df[df['col']>value]`	boolean indexing

Method	Description
`df.query('col>value')`	query-style filtering

4️⃣ Data Cleaning

Method	Description
`df.dropna()`	missing values drop
`df.fillna(value)`	missing values fill
`df.isna()`	check missing values
`df.duplicated()`	check duplicates
`df.drop_duplicates()`	drop duplicate rows
`df.replace(old,new)`	replace values

5️⃣ Data Manipulation

Method	Description
`df.sort_values(by='col')`	sort by column
`df.sort_index()`	sort by index
`df.rename(columns={'old':'new'})`	rename columns
`df.set_index('col')`	set column as index
`df.reset_index()`	reset index to default
`df.drop(columns='col')`	drop column
`df.astype(dtype)`	change column dtype

6️⃣ Aggregation & Grouping

Method	Description
`df.groupby('col')`	group by column
`df.groupby('col').sum()`	grouped sum
`df.groupby('col').mean()`	grouped mean
`df.pivot_table(index='col1', columns='col2', values='col3', aggfunc='mean')`	pivot table

7️⃣ Merging & Joining

Method	Description
`pd.concat([df1, df2])`	concatenate DataFrames
`pd.merge(df1, df2, on='col', how='inner')`	merge DataFrames
`df.join(df2, on='col')`	join DataFrames

8️⃣ Time Series

Method	Description
`pd.to_datetime(df['col'])`	convert to datetime
`df.set_index('date_col')`	set datetime index
`df.resample('M').sum()`	resample time series
`df.shift(1)`	lag/lead data

9️⃣ File Handling

Method	Description
`df.to_csv(filename)`	save to CSV
`df.to_excel(filename)`	save to Excel
`df.to_json(filename)`	save to JSON
`df.to_sql(table, conn)`	save to SQL

🔟 Visualization (with Pandas)

Method	Description
`df.plot()`	line plot
`df.plot(kind='bar')`	bar plot
`df.plot(kind='hist')`	histogram
`df.plot(kind='box')`	box plot
`df.plot(kind='scatter', x='col1', y='col2')`	scatter plot

💡 Example

```
import pandas as pd

data = {'Name':['Alice','Bob','Charlie'], 'Age':[25,30,35], 'Salary':[50000,
60000,70000]}
df = pd.DataFrame(data)

print(df.head())
print(df['Age'])
print(df[df['Salary']>55000])
df['Age'] = df['Age'] + 1
print(df)
```